GENERAL

Typically 6 to 12 weeks after receipt of a unit order, Gas Turbine Requisition Engineering will issue a package of drawings and specifications commonly called "Customer Drawings," "Customer Design Documents," or "Customer Interface Drawings." These documents are listed by Model List Item (MLI) number in two groups for typical MS6001B/MS7001EA gas turbine-generator units. See Table 3 of EM 5016, Installation Documents for MS6001B Gas Turbines.

Standard documents are documents which are "always" supplied and are contractually committed to be supplied.

Supplementary documents are drawings which are not optional, but are not part of the contractually committed items. Depending upon the specific unit configuration, some or all of these documents may be supplied.

STANDARD "CUSTOMER DRAWINGS" CONTENT SUMMARY

MLI 0301 – Outline, Gas Turbine Package Conn-Electrical

- Locates junction boxes, receptacles, and electrical devices requiring field interconnection.
- Locates grounding studs.
- Suggests cable routing through embedded conduit (MLI 0323) and entry points.
- Used in conjunction with Foundation Interface Outline (MLI 0323) and Diagram, Block, Cable (MLI 0463).

MLI 0306 – Outline, Mechanical – Gas Turbine and Load

- Provides overall Gas Turbine-Generator (GT-G) unit dimensions and arrangement.
- Provides reference dimensions for all centerline equipment packages.
- Provides dimensions of Gas Turbine-Generator supplied equipment which extends from the centerline packages.
- Provides space removal requirements for maintenance purposes (e.g., generator rotor, tube bundles, etc.).

MLI 0313 – Outline, Gas Turbine Package Conn-Piping

- Provides location for all BOP/field run piping connections to turbine package.
- Locates all drain outlet points.
- Used in conjunction with Notes, Gas Turbine Package Connection Outline-Piping (MLI 0314), and various Piping Schematics (MLI 04XX).

MLI 0314 – Notes, Gas Turbine Package Conn Outline-Piping

- Defines all Gas Turbine piping connections located in Outline, Gas Turbine Package Conn-Piping (MLI 0313).
- Provides connection interface data (e.g., thread, flange size, rating, etc.).
- Identifies drains which are normally plugged or connected to station sump.

MLI 0323 – Foundation Interface Outline

- "A" size drawing which lists drawings needed to define the following parameters for the accessory compartment, gas turbine, generator, skid and barrier foundations.
    1. Size and location of the support pads or sole plates of the various compartments.
    2. Static loading on pads.
    3. Recommended conduit embedments.
    4. Suggested design for embedments to which centerline guides are fastened.
    5. Suggested design for jacking posts.
    6. Dynamic load considerations ("COMDOC" Notes).
    7. Recommended size and height (above machine baseline) for foundation anchor bolts.
- Used in conjunction with Foundation Bolting Arrangement (MLI 1603) and Cable Routing (MLI 0301).

MLI 0326 – Outline, Foundation Interface, Off-Base Mechanical

- "A" size drawing which lists drawings and notes needed to define the following parameters for Gas Turbine supplied off-base skids.

    1. Foundation interface and design parameters for "mechanical" off-base equipment.

    2. Envelop dimensions of skid, including door location(s) (if lagged) to avoid interference with other equipment.

    3. Piping electrical and ground connection points.

    4. Equipment weight, Center of Gravity and pad loading.

- Used in conjunction with Diagrams, Schematic Piping-Various (MLI 04XX).

MLI 0330 – Outline, Foundation Interface, Off-Base Electrical

- "A" size drawing which lists drawings needed to define the following parameters for Gas Turbine supplied off-base electrical equipment.

    5. Foundation interface and design parameters for electrical off-base equipment.

    6. Envelope dimensions and door locations (if appropriate).

    7. Electrical connections details (conduit entry locations for underground cable if required).

    8. Equipment weight, Center of Gravity and pad loading.

- Used in conjunction with Diagrams, Schematic Piping-Various (MLI 04XX).

MLI 0408 – Weight and Center of Gravity (CG), Gas Turbine Packages

- Provides individual weight and CG for major packages of equipment.

- Provides suggested lifting arrangements.

MLI 0414 – Device Summary

- Provides control system components summary.
- Supplies functional characteristics of all mechanical and electrical devices.
- Provides device condition code.
- Lists electrical devices by IEEE electric device nomenclature code.
- Used in conjunction with Diagrams, Schematic Piping-Various (MLI 04XX), Diagram, Block, Cable (MLI 0463) and Diagram, One Line (MLI 0444).

MLI 04XX – Diagrams, Schematic Piping-Various

- Portrays functional relationship of mechanical and electrical devices located within each Turbine-Generator system.
- Identifies device nomenclature, location and device settings.
- Identifies drawing number for all applicable piping schematics.
- Provides flow rates, pressure and temperature requirements at BOP/field run connection points.
- Documents system capacities and liquid levels.

MLI 0438 – List, Standard Customer Drawings

- "A" size drawing which lists documents that provide the following general and supplemental information.

  a. Commonly required specifications and recommendations for selection of cooling water, fuels, lube oil, etc.

  b. Design parameters for sizing and locating embedded conduits for interconnecting cables.

  c. Information to interpret piping and welding symbols.

  d. Information on equipment preservation requirements and final paint.

  e. Insulation recommendations.

- Accompanying "E" size drawing provides Overview of Customer Drawings supplied for the Gas Turbine-Generator unit (always 935E669).

MLI 0444 – Diagram, One Line

- Simplified electrical schematic of power system from generator ground through high voltage bus.
- Defines protective relaying, excitation and synchronizing systems.
- Defines auxiliary power system requirements, including types and sizes of motor starters.
- Schematic display of distribution panels and functional display of cable remote control panel.
- Uses IEEE electrical device nomenclature code.

MLI 0463 – Diagram, Block, Cable

- Lists cable and wire requirements interconnecting Gas Turbine supplied equipment.
- Lists cable size and voltage level requirements for Gas Turbine supplied cables.
- Identifies cables and wire not supplied by Gas Turbine, and provides design parameters for customer guidance.
- Used in conjunction with Outline, Gas Turbine Package Conn-Electrical (MLI 0301), and Foundation Interface Outline (MLI 0323).

MLI 1603 – Foundation Bolting Arrangement

- Defines bolting arrangement of the Gas Turbine package to the main foundation.
- Defines materials (shims, keys and keyways) required for leveling, aligning, keying and bolting centerline packages.
- Identifies materials supplied and not supplied by Gas Turbine.
- Supplements information supplied in the Foundation Interface Outline (MLI 0323).

Note: The terms "supplements" or "used in conjunction with" imply a primary relationship, meaning that information contained in these documents needs to be used and correlated.

NOTES AND CAUTIONS ON USE

Because of a number of factors including lack of familiarity of users, changing product arrangements and packaging, special applications, etc., problems have occurred in using these documents. A combination of experience and specific drawing study is needed to fully interpret the engineering and construction implications.

The following are common areas of confusion.

- Most of the drawings include a mixture of information, and a specific drawing may need to be referenced by mechanical, electrical and civil/structural engineering groups.
- Due to the present use of supplemental documents, many drawings may need to be referenced in order to obtain complete information about a specific area.
- Differences exist in scope of supply. For example, Gas Turbine supplies pipe and electrical cable for some of the off-base equipment (e.g., Exhaust Frame Blowers, Demister and Gas Fuel Module), but supplies only the skid for most off-base systems.
- Foundation design, foundation embedments, and sub-sole plates are usually the responsibility of others, with Gas Turbine supplying only interface requirements and specific recommendations.
- Although Gas Turbine supplies shim packs for support of the various packages, the sizing and procurement of sub-sole plates (as well as all foundation embedments) is usually the responsibility of others. It should be noted that dimensions given in the details of the Foundation Interface Outline (MLI 0323) are for the pads or sole plates of the equipment. The sub-sole plates may need to be larger, or can be smaller, depending on the size of shims, attachment of axial positioning block, etc.
- Although Gas Turbine recommends the size (diameter), height above sub-sole plate, and material of the anchor bolts, the remainder of the bolt design and the procurement is usually the responsibility of others.

- Drawings listed under MLI 0323 sometimes carry the MLI 0323 identity in the title block, which can mislead a user into thinking that all of MLI 0323 is represented in a single drawing. The "A" size "top drawing" must be referenced to determine the full list of required drawings.

- In some cases, Gas Turbine supplies the electrical cables but does not define skid location or conduit runs. This requires feedback to Gas Turbine during the requisition phase to allow required sizing of the length for these cables.

- Although Gas Turbine does not locate many of the off-base skids, there are design constraints which must be considered in the final equipment arrangement (e.g., distance of skid from unit pipe connections, distance from nearby obstructions, etc.).

- Drawings with "suggested" or "recommended" information should be regarded with extra caution, as sometimes the "Max. Case" is represented which may be excessive and costly for specific unit needs (e.g., conduit runs in the main foundation).

- Gas Turbine uses a more conservative fill rate on conduits (e.g., a 25% fill of a 4-inch conduit as maximum vs. the industry standard of 40% fill).

- Although not noted on the drawings, additional "straight through" conduit runs should not be above the recommended due to heat from the exhaust plenum.

- Drawings listed under MLI 0326 may have their own separate and different MLI Numbers.

- Although titled as foundation interface drawings, both MLI 0326 and MLI 0330 are lists of skid and module drawings that contain more information than just foundation interface.

- Piping schematics contain more than just piping or fluid flow information (e.g., relative location of instruments, etc.), and should be regarded as system schematics.

Note: The term "off-base" is used to mean support separate from the main centerline foundation.